

EXP 1

AIM:

To perform the required signal processing analysis and obtain the output using MATLAB.

Software Used:

MATLAB (MATrix LABoratory)

CODE:

```
clc;
clear;
close all;

% Plant parameters
K = 5;
T = 10;
Td = 2;          % Time delay = 2 seconds

% Plant without delay
G0 = tf(K,[T 1]);

% 1st-order Pade approximation
[num1,den1] = pade(Td,1);
Pade1 = tf(num1,den1);
G1 = G0 * Pade1;

% 2nd-order Pade approximation
[num2,den2] = pade(Td,2);
Pade2 = tf(num2,den2);
G2 = G0 * Pade2;

% Display transfer functions
disp('Plant without delay:');
G0

disp('Plant with 1st-order Pade:');
```

G1

```
disp('Plant with 2nd-order Pade:');
```

G2

```
% Time vector and unit-step input
```

```
t = 0:0.01:60;
```

```
u = ones(size(t));
```

```
% Simulate using lsim()
```

```
y0 = lsim(G0,u,t);
```

```
y1 = lsim(G1,u,t);
```

```
y2 = lsim(G2,u,t);
```

```
% Plot all responses
```

```
figure;
```

```
plot(t,y0,'LineWidth',1.5);
```

```
hold on;
```

```
plot(t,y1,'--','LineWidth',1.5);
```

```
plot(t,y2,':','LineWidth',1.5);
```

```
grid on;
```

```
xlabel('Time (s)');
```

```
ylabel('Output');
```

```
title('Comparison of Step Responses');
```

```
legend('No Delay','1st-Order Pade','2nd-Order Pade');
```

```
hold off;
```

```
% Calculate step-response characteristics
```

```
info0 = stepinfo(y0,t);
```

```
info1 = stepinfo(y1,t);
```

```
info2 = stepinfo(y2,t);
```

```
% Display results using fprintf()
```

```
fprintf('\n=====\\n');
```

```
fprintf(' STEP RESPONSE PERFORMANCE COMPARISON\\n');
```

```
fprintf('=====\\n');
```

```
fprintf('%-20s %-15s %-15s\\n', ...
```

```
    'System','Peak Overshoot (%)','Settling Time (s)');
```

```
fprintf('-----\\n');
```

```

fprintf('%-20s %-15.2f %-15.2f\n', ...
        'No Delay',info0.Overshoot,info0.SettlingTime);

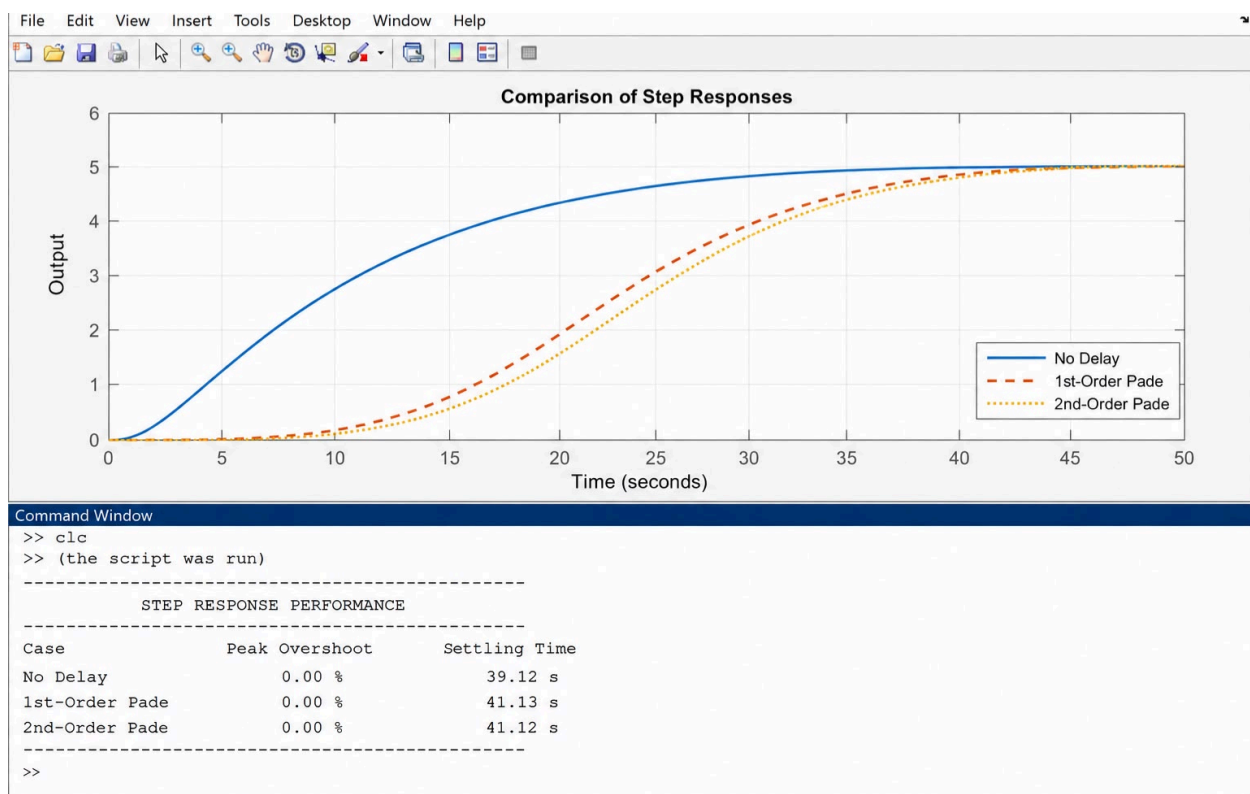
fprintf('%-20s %-15.2f %-15.2f\n', ...
        '1st-Order Pade',info1.Overshoot,info1.SettlingTime);

fprintf('%-20s %-15.2f %-15.2f\n', ...
        '2nd-Order Pade',info2.Overshoot,info2.SettlingTime);

fprintf('=====\n');

```

OUTPUT:



EXP 2

Aim:

To analyze the stability of the plant using a PD controller in Scilab by forming the closed-loop transfer function, plotting the Nyquist diagram, and determining the phase margin and gain margin.

Software Used:

MATLAB (MATrix LABoratory)

CODE:

```
clc;
clear;
clf;

// Define Laplace variable
s = %s;

// Plant
G = syslin('c', 1, s^2 + s);

// PD Controller
Kp = 10;
Kd = 2;
C = syslin('c', Kp + Kd*s, 1);

// Open-loop transfer function
L = C * G;

// Closed-loop transfer function using feedback()
T = feedback(L, 1);

// Display transfer functions
disp("Plant G(s) = ");
disp(G);

disp("PD Controller C(s) = ");
```

```
disp(C);

disp("Open-loop transfer function L(s) = ");
disp(L);

disp("Closed-loop transfer function T(s) = ");
disp(T);

// Nyquist plot
scf(1);
nyquist(L);
xtitle("Nyquist Plot of Open-Loop System");

// Phase margin
[PM, fpm] = p_margin(L);

disp("Phase Margin (degree) = ");
disp(PM);

disp("Phase crossover frequency (Hz) = ");
disp(fpm);

// Gain margin
[GM, fgm] = g_margin(L);

disp("Gain Margin (dB) = ");
disp(GM);

disp("Gain crossover frequency (Hz) = ");
disp(fgm);

// Closed-loop poles
disp("Closed-loop poles = ");
disp(roots(s^2 + 3*s + 10));
```

OUTPUT:

